MetroAI: availability and PodDisruptionBudget

Group 2 | Milestone 5

Without a PodDisruptionBudget (PDB), concurrent voluntary evictions during node maintenance can remove every ready application replica. The load balancer then has no healthy destination, causing intermittent 5xx responses until replacement pods become ready. Terminating backends can also interrupt in-flight requests. Missing a PDB does not itself cause an error; a disruption must remove too much serving capacity.

MetroAI uses at least two GKE replicas and a PDB with minAvailable: 1, limiting supported voluntary evictions to preserve one available replica. Rolling updates are separately configured with maxUnavailable: 0 and maxSurge: 1. Readiness queries PostgreSQL transit data and returns 503 when unavailable; application-only liveness avoids database-triggered restart loops. The load balancer checks readiness too. Connection draining, a pre-stop delay and a termination grace period reduce interruption while endpoints are removed.

Our bounded HTTP test triggered actual HPA scaling from two to three and then four replicas; all 35,715 responses were HTTP 200. A separate deletion rehearsal showed automatic pod replacement. These observations do not prove the PDB prevented an outage: direct pod deletion bypasses PDBs, and PDBs do not govern Deployment rolling updates or prevent node crashes. The live deletion demo demonstrates controller replacement, not PDB enforcement. We did not deliberately remove all serving capacity or disable the PDB to manufacture errors. The existing zonal database, shared dependencies and pods placed on one node remain availability limits even with multiple replicas.

Sources: https://kubernetes.io/docs/concepts/workloads/pods/disruptions/
https://docs.cloud.google.com/kubernetes-engine/docs/how-to/ingress-configuration